

IR thermometers are reliable and very useful for single-spot temperature readings. However, when scanning large areas or components, these may miss critical components

that are nearing failure and need repair. Thermal imaging cameras, on the other hand, scan entire components, giving you instant diagnostic insights with the full extent

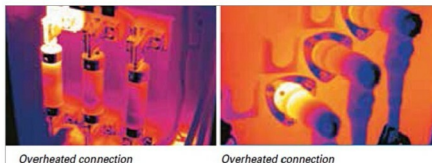


Fig. 4: Inspection of a substation revealing overheated connections (Courtesy: www.bsria.co.uk)

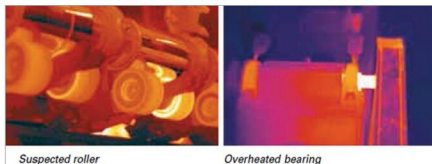


Fig. 5: Suspected roller and overheated bearing (Courtesy: www.bsria.co.uk)



Fig. 6: (L) IR thermometer and (R) thermal imaging camera (Courtesy: www.bsria.co.uk)

of problems. These are easier to use and find problems faster with higher accuracy.

How thermal imaging cameras work

The hotter the object, the more its infrared radiation. Infrared energy coming from the object is focused by the optics onto an infrared detector. The detector sends this information to a sensor for image processing using complex calculations. The sensor translates the data into an

image that can be viewed on the LCD screen. Each pixel of the image is a measure of the temperature at different points. This is achieved by complex algorithms performed in the thermal imaging camera. If the resolution of an image by a camera is 60x60 pixels, it is equivalent to the use of 3600 IR thermometers at the same time.

Choosing the best

Thermal imaging cameras are a long-term investment. The safety of entire installations and operating people relies on them. So you need to choose a camera that serves your purpose over a long period of time. The things you should look for are hardware, software, accessories (lens and display), after-sales services and training.

As regards their use, you may face a few challenges. Honeywell experts explain these as follows: Thermal properties of materials depend on their compositions. In contrast to insulators, metals heat up quickly. Differences in thermal conductivity can lead to large temperature differences under certain circumstances. So always check the thermal conductivity of the material and its surroundings before inspection.

Emissivity is another important factor that is highly dependent on material properties. It is the efficiency with which an object emits infrared radiation. You should set the right emissivity. Else, temperature measurements will be incorrect.

Some materials have mirror-like reflection properties, especially if these are polished or non-oxidised metals. These reflections can lead to misinterpretation of the thermal image. Therefore choose the right angle to avoid such reflections.

Ambient temperatures can mislead the actual temperature readings. High ambient temperature may increase temperature of the entire object, thus masking the actual hot spots. Low ambient temperature may cool down the hot spots.

The right combination for the best detection is high resolution, low thermal sensitivity, high accuracy and LED lights in your camera. **EFY**



If you look at the thermal image you might think that the gold paint is colder than the mug surface. In reality they have exactly the same temperature, the difference in intensity of infrared radiation is caused by a difference in emissivity.

Fig. 7: Difference in emissivity causes difference in intensity of radiation (Courtesy: www.bsria.co.uk)